

AMMAD: A Multimodal Approach for Disease Diagnosis with Symptoms and Tests

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Abstract. Accurate and timely disease diagnosis is a cornerstone of effective medical treatment, yet it remains a complex and challenging task, particularly for less experienced healthcare professionals. This paper introduces a novel system for disease diagnosis by integrating clinical symptoms with test results, aiming to support healthcare providers in making more informed and reliable decisions. The dataset used in this study comprises 66,508 records collected from hospital’s software in healthcare facilities, which includes both clinical symptom data and laboratory test results. A neural network model is proposed to process and combine these multi-modal inputs. Our model obtains a high accuracy of 74.49% and an F1 score of 65% when tested on 13,302 data entries. Additionally, we implement a Web Service API in real-world applications, offering real-time diagnostic assistance and enhancing the overall efficiency of healthcare delivery. This approach contributes to more effective patient care by improving diagnostic accuracy and decision-making in clinical settings.

Keywords: Deep learning · Disease Diagnosis · Multi-Modal · Clinical Application

1 Introduction

In modern healthcare, disease diagnosis [10] is central to effective patient care and directly shapes treatment decisions. Accurate diagnosis from clinical symptoms and *laboratory test results* [14] is essential, as early detection and precision markedly improve treatment outcomes. Yet, this process remains complex, particularly given the large volumes of patient data and the demand for rapid decision-making in high-pressure settings. Despite advances in medical technology [5], challenges persist in integrating and analyzing heterogeneous data sources, such as clinical symptoms and diagnostic laboratory results [7], to achieve more reliable diagnoses. These difficulties are especially acute for less experienced healthcare providers, including newly graduated doctors, who may struggle to interpret such complex data effectively [3].

A major challenge in diagnostic model development is the quality and consistency of clinical data [17]. Symptom descriptions are often subjective, while laboratory results may vary due to equipment, methodology, and human factors, leading to unreliable inputs [15]. Missing or incomplete records further compromise predictive accuracy. Integrating multimodal data [8], such as textual symptoms and numerical test results, remains particularly difficult, as each modality must be fused into a unified framework without loss of critical information [18]. Although deep learning performs well on single data sources, multimodal fusion is still complex and underexplored. The “black-box” nature of deep models [13] also limits interpretability and clinician trust. Prior studies have addressed aspects of disease diagnosis [4, 11]: NLP models such as BERT [9] process clinical text effectively, while CNNs [6, 16] excel in medical imaging. Yet most approaches remain unimodal, overlooking the benefits of combining text, numerical, and imaging data. Emerging frameworks such as DeepMed [19] and DeepHealth [1] attempt this integration but still face challenges in seamless fusion, computational efficiency, and real-time clinical use.

In this paper, we introduce a novel approach to disease diagnosis through our Symptom and Test Result Integrated MLP models [2, 12], which tackles the complex challenge of multimodal integration by leveraging both clinical symptoms and laboratory test results. Our primary focus is on improving diagnostic accuracy while ensuring computational efficiency. We present a comprehensive methodology that involves two distinct stages: (1) Group Classification and (2) Disease Classification. Our approach is further enhanced by experimenting with three key strategies: (i) Predicting the disease group using clinical symptom data first, followed by predicting the specific disease based on test result data; (ii) Predicting the disease group using test result data, then predicting the specific disease based on clinical symptom data. (iii) Using both clinical symptom and test result data as separate, discrete inputs, followed by prediction based on ICD-10 codes and the top 5 most accurate disease codes.

2 Background

2.1 Classification for Disease Diagnosis

Definition 1 (Disease Classification). *Let x be a feature vector representing patient data, such as clinical symptoms or laboratory test results. A disease classification model based on deep learning is a function f^μ that maps x to an output $y = f^\mu(x)$, where y denotes either a categorical label or a probability distribution over the class set Y . The final prediction assigns x to class Y_j , corresponding to the highest probability in y . Here, μ specifies the underlying model architecture (e.g., neural network, decision tree).*

Example 1. Let $Y = \{Y_1, Y_2, \dots, Y_n\}$ denote the set of possible disease categories, and let $\mu \in \{\text{MLP, SVM, Random Forest, } \dots\}$ denote the choice of classification model. Given an input vector x_1 representing clinical symptoms and laboratory test results, applying the model $\mu = \text{MLP}$ yields $Y_k = f^{\text{MLP}}(x_1)$, where $Y_k \in Y$ is the predicted disease category.

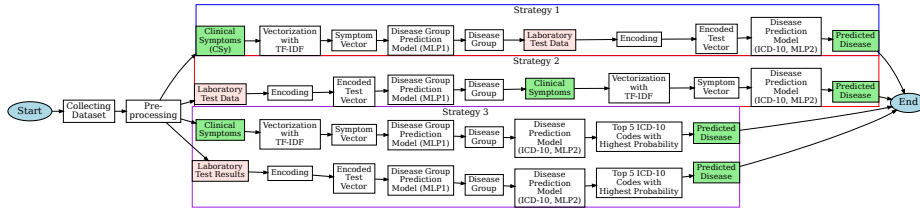


Fig. 1: Strategies for Disease Diagnosis.

2.2 Dataset of Diseases

The dataset was collected from VNPT-HIS software deployed in healthcare facilities across Soc Trang province (Vietnam) between 2022 and 2023, adhering to data security regulations. It includes patient records focusing on clinical symptoms and laboratory test results, with disease codes standardized under WHO’s ICD-10 and limited to common diagnoses such as digestive, respiratory, and circulatory diseases. The raw data contained diverse formats (e.g., textual lab results and clinical symptoms) with spelling errors, abbreviations, and noise. Preprocessing involved converting categorical test results into numerical values, handling missing data, and cleaning clinical symptom text (removing special characters, lowercasing, tokenization). Weka¹ was used for attribute selection, while Word2Vec represented symptoms as vectors to capture semantic relationships. The final dataset comprises 66,508 records with 199 attributes, providing a robust foundation for deep learning-based diagnostic modeling.

3 Strategies for AMMAD

The disease prediction framework integrates three complementary strategies, clinical symptoms, laboratory test results, and ICD-10 classifications, to enhance diagnostic accuracy and support reliable decision-making (Figure 1). (1) *Clinical symptoms*: Raw symptoms are vectorized using TF-IDF to emphasize key features while reducing noise. These vectors are processed by MLP1 to predict disease groups, enabling rapid and accurate grouping, particularly useful for less experienced practitioners. (2) *Laboratory test results*: Heterogeneous test data are standardized into vectors and analyzed by MLP1 to predict disease groups. This strategy complements symptom-based predictions by bridging clinical evidence with laboratory findings, thereby increasing diagnostic certainty. (3) *ICD-10 integration*: TF-IDF symptom vectors are classified by MLP2 into ICD-10 codes, ensuring compliance with international standards. The model outputs the top five most probable codes, providing clinicians with multiple diagnostic options for further testing or treatment planning.

¹ <https://pypi.org/project/python-weka-wrapper3/>

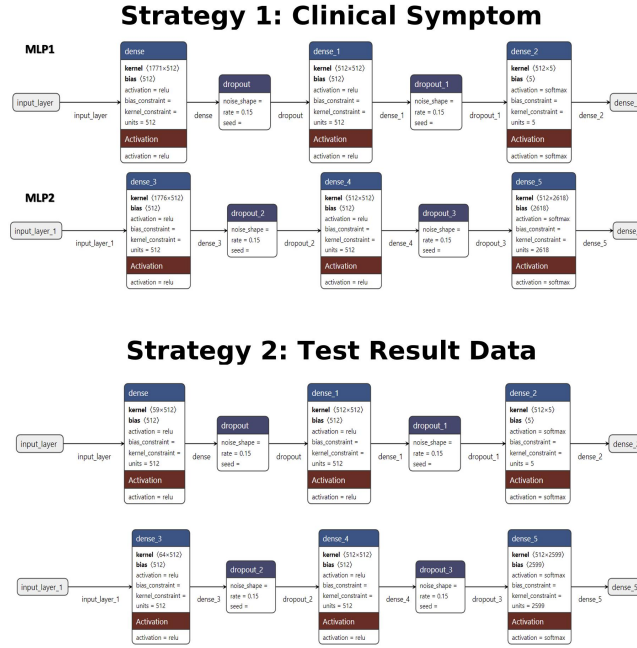


Fig. 2: The MLP1 and MLP2 for Clinical Symptom and Test Result Data.

4 Proposed MLP Models

4.1 Models For Clinical Symptom

The MLP model for clinical symptom data classifies disease groups by capturing complex, non-linear relations between symptoms and diagnoses. Input vectors, preprocessed with TF-IDF, are fed into a dense layer with ReLU activation, enabling the network to learn intricate patterns. Dropout layers (*dropout₁*, *dropout₂*, *dropout₃*) are inserted to reduce overfitting by randomly deactivating neurons during training. Subsequent dense layers (*dense₂*–*dense₄*) extract progressively abstract features, refining the representation of symptom data. The final dense layer (*dense₅*) applies softmax activation to generate a probability distribution across disease categories, with the highest probability indicating the predicted disease. By combining ReLU-based learning with softmax classification, the model achieves robust performance in handling the complex symptom–disease mappings essential for medical applications.

4.2 Models for Laboratory Test Result Data

The MLP model for test result data (Figure 2) is designed to process diverse laboratory diagnostics, including both numerical values and categorical outcomes.

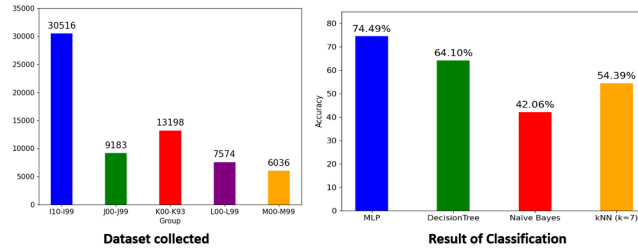


Fig. 3: The collected dataset (right side) and experimental result.

Input vectors pass through an initial dense layer with ReLU activation, enabling the network to capture complex patterns. Subsequent dense layers refine these representations, while dropout layers ($dropout_1$, $dropout_2$, $dropout_3$) deactivate 15% of neurons during training to prevent overfitting and improve generalization. The final dense layer ($dense_5$) applies a softmax function to produce a probability distribution over 2,599 ICD-10 disease categories, ensuring mutually exclusive predictions, with the highest probability.

5 Experiments and Discussion

5.1 Dataset and Configurations

The dataset, illustrated in Figure 3, is divided into five disease groups according to ICD-10. The largest group, I10–I99 (cardiovascular and circulatory diseases), contains 30,516 samples, followed by K00–K93 (digestive system diseases) with 13,198 samples, J00–J99 (respiratory disorders) with 9,183 samples, L00–L99 (skin and subcutaneous tissue diseases) with 7,574 samples, and M00–M99 (musculoskeletal and connective tissue diseases) with 6,036 samples. Overall, 53,206 samples are allocated for training and 13,302 for testing, ensuring reliable performance evaluation across multiple categories. The dataset’s diversity provides a strong foundation for developing and validating disease classification models. For implementation, we employ TensorFlow (v2.16.0), Keras (v2.16.0), Gensim (v4.3.3), Pandas (v2.2.2), and Scikit-learn (v1.6.1). Experiments are conducted on a system equipped with an Intel Core i7-11800 (2.30 GHz), 16 GB RAM, and Windows 11 Pro OS. The dataset and implementation are available on GitHub².

5.2 Experimental Results

Result of Strategies: We compare three diagnostic strategies using Multi-Layer Perceptron (MLP) models in terms of accuracy, F1-score, and computation time. Strategy 3, which combines clinical symptoms with ICD-10 classifications, achieved the highest accuracy (74.49%) and F1-score (65.14%) but required the

² <https://github.com/truongthanhma/AMMAD/>

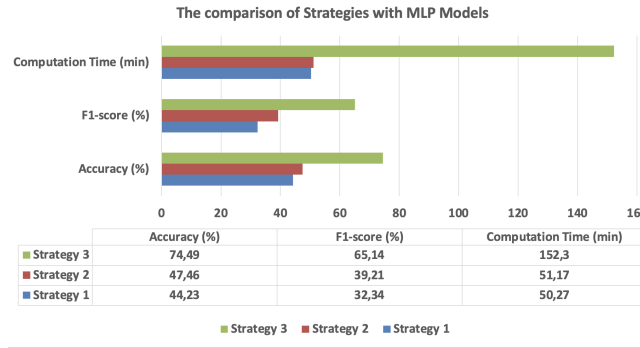


Fig. 4: A comparison of Strategies in F1, Accuracy and Computation Time.

longest computation time (152.3 minutes). This suggests that ICD-10 integration improves prediction quality but increases computational complexity, making it suitable when accuracy outweighs efficiency. Strategy 2, based on laboratory test results, delivered moderate accuracy (47.46%) and F1-score (39.21%) with 51.17 minutes of computation, representing a balance between performance and efficiency. Strategy 1, relying only on clinical symptoms, yielded the lowest accuracy (44.23%) and F1-score (32.34%) but completed fastest (50.27 minutes), reflecting the limitations of a reduced feature set. Overall, the results highlight a trade-off: Strategy 3 offers the highest precision but is computationally expensive, Strategy 2 provides a practical middle ground, and Strategy 1 is efficient but less reliable.

Comparison between MLP and Traditional MLs We compare four machine learning models—MLP (Multi-Layer Perceptron), Decision Tree, Naïve Bayes, and k-Nearest Neighbors (kNN)—for disease classification using clinical symptoms and laboratory test results (Figure 5). The MLP achieves the highest accuracy (74.49%), reflecting its capacity to capture complex, non-linear relationships in medical data, and proving most robust for this task. The Decision Tree attains 64.10% accuracy, offering interpretability with moderate performance, though limited by dataset complexity and non-linearity. Naïve Bayes records the lowest accuracy (42.06%), as its assumption of feature independence fails in real-world clinical contexts. The kNN model achieves 54.39%, performing moderately but hindered by the “curse of dimensionality”, which reduces its effectiveness with high-dimensional data.

5.3 Interface and Discussion

The STID system interface provides an efficient and user-friendly platform for disease diagnosis based on clinical symptoms and laboratory test results. Developed with Python and Flask-API, it operates through a POST method that accepts JSON-formatted input, processes it with predictive models, and returns

The screenshot shows a web application interface for disease diagnosis. It features a header with navigation links: "NHẬP THÔNG TIN BỆNH NHÂN", "THỰC CHẨN LÂM SÀNG", "SINH HÓA", "HUYẾT HỌC", "NƯỚC TIỂU", "M. SINH", and "KHÁC". The main area is divided into three sections: 1. A large input area for entering symptoms and test results, 2. A section for displaying the diagnosis, and 3. A "SUBMIT" button. The interface is clean and professional, with a light gray background and red borders around the input and output areas.

Fig. 5: Application of Disease Diagnosis.

the corresponding ICD-10 disease code. Interaction with the system is seamless. Users enter symptoms and test results, submit the data, and receive instant diagnostic outputs on the interface. This ensures accessibility for healthcare professionals and non-specialists alike. The system offers several advantages. Its simple interface enables rapid data entry, while Flask-API ensures efficient processing and timely feedback, critical for clinical decision-making. Alignment with ICD-10 guarantees internationally standardized diagnoses, and real-time predictions enhance treatment planning.

6 Conclusion and future work

This study developed a decision support system for disease diagnosis based on clinical symptoms and laboratory test results. Leveraging deep learning and multi-layer neural networks, data from VNPT HIS were preprocessed using feature selection (InfoGain), normalization, and class-balancing techniques. The trained models were deployed as a web-based API to provide timely diagnostic support and improve patient outcomes. Future work should expand the dataset with additional symptoms and test results to build a more comprehensive clinical dictionary. Refining laboratory values into finer categories could enhance predictive accuracy, while external validation would improve reliability. Developing multi-class models may enable detection of comorbidities and secondary diseases, and integrating imaging and biopsy data would provide richer clinical evidence, further strengthening the system's practical effectiveness.

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